



An agentic satellite query engine

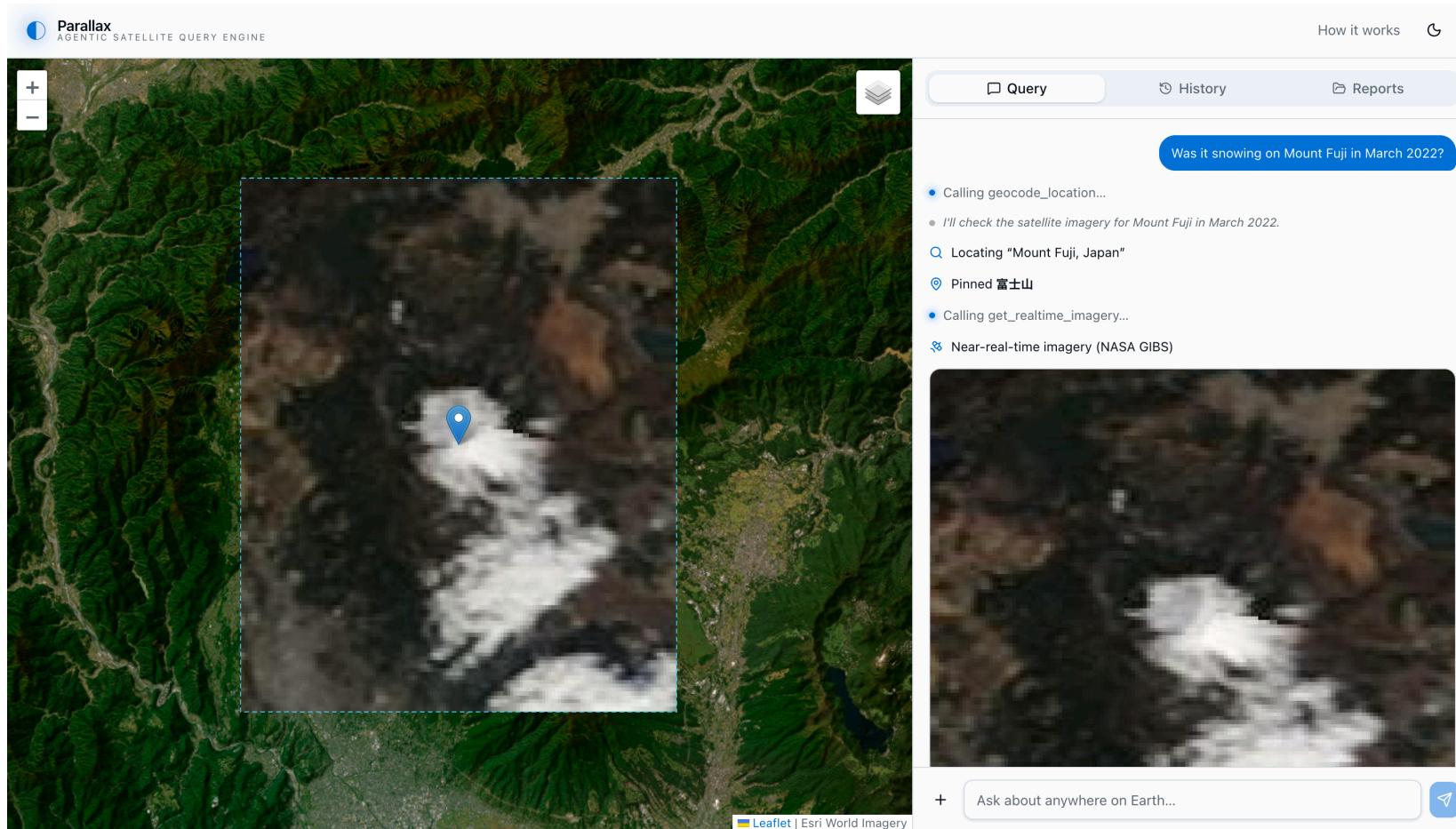
Ask a plain-language question about anywhere on Earth. A LLM agent locates the place, pulls live satellite data, reads the actual pixels, and answers — grounded in what the imagery shows.

CS 153 · Frontier Systems — Final Project

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Overview

Parallax turns a natural-language question into a grounded, evidence-based answer about a place on Earth. Under the hood it's an agent: instead of answering from memory, it uses tools to fetch live satellite imagery and reasons over the real pixels.



Everything is built on free, open data (NASA, ESA/Copernicus, Esri, OpenStreetMap). The only paid dependency is the Claude API that runs the agent.

Why I built it



Satellite data is open — but locked behind expertise

Agencies publish fresh global imagery for free, yet interpreting it normally takes a GIS / remote-sensing specialist: projections, band math, tiling, a dozen APIs. Most people can't get an answer out of it.



Ordinary people have place-based questions

"Has it snowed at Tahoe before my ski trip?" "Is that lake still full?" "Is there a fire near my town?" — real questions with real imagery to answer them, and no simple way to ask.

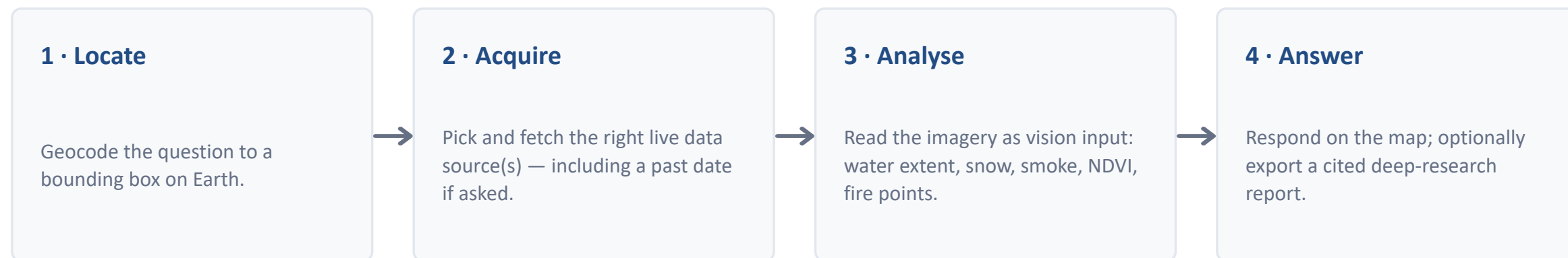


It's newly feasible — and genuinely useful

Agentic tool use plus vision-capable models means a single agent can now locate, fetch, and read imagery end-to-end. The interesting part was seeing how far that actually goes.

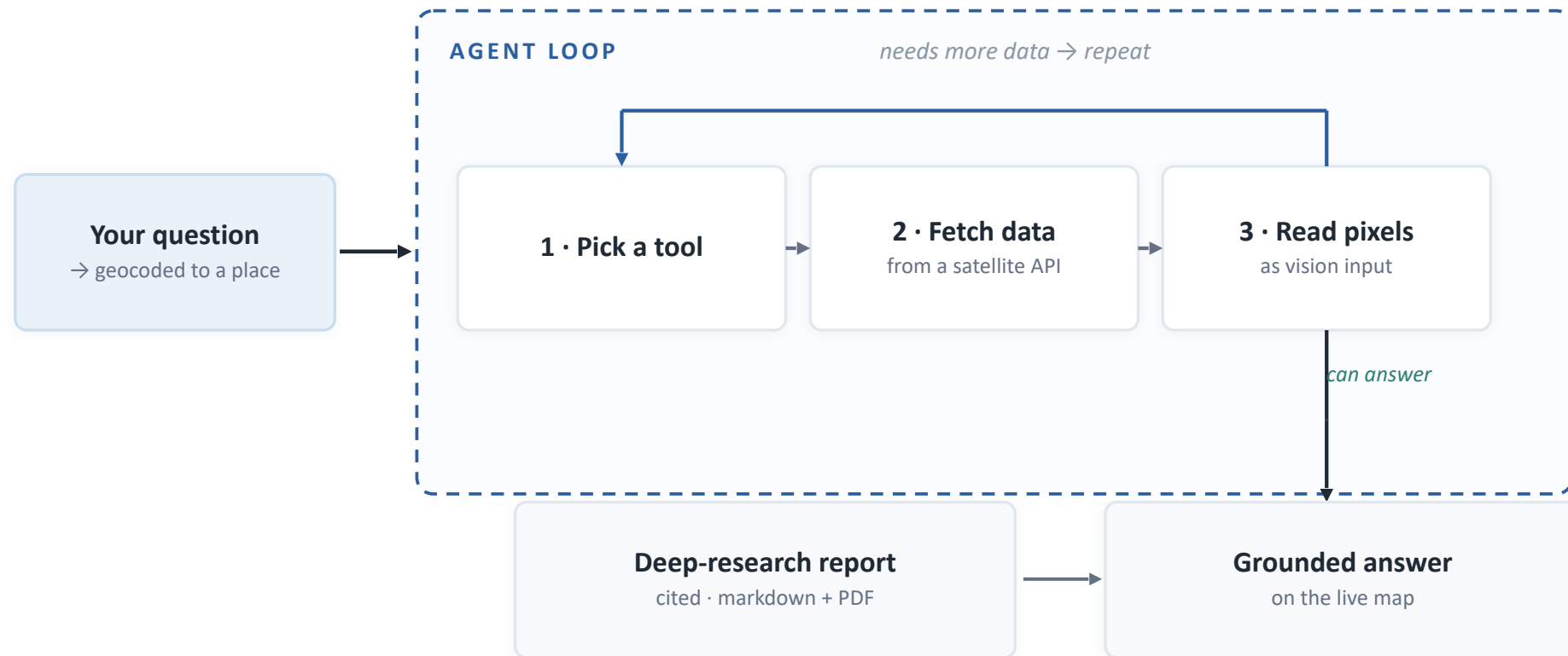
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The agent loop

Rather than one shot, the agent runs a loop: pick a tool, fetch data, look at the result, and decide whether it can answer or needs more.



In practice

- Geocode runs first, then the loop.
- The model chooses tools per question.
- Often 2–3 tools combined.
- Each step streams live to the UI.
- Capped at 8 turns to stay bounded.

The agent's tools

Seven tools; the model picks which to call for each question.

Tool	What it does
<code>geocode_location</code>	Resolve a place name to lat/long + bounding box (called first)
<code>get_active_fires</code>	Active-fire detections — exact counts and locations
<code>get_realtime_imagery</code>	Daily wide-area imagery + the full historical archive (any date)
<code>get_highres_imagery</code>	~10 m true-color detail: cities, water bodies, coastlines
<code>get_basemap_imagery</code>	Sub-meter detail: buildings, runways, ships, stadiums
<code>get_vegetation_index</code>	NDVI — crop health / drought (mean/min/max + % healthy)
<code>compare_over_time</code>	Change detection between two dates

The model frequently combines tools — e.g. fire detections plus wide-area smoke imagery, or NDVI plus change-detection over two dates.

Where the data comes from

Five open sources, each with different resolution and freshness — the agent trades them off per question.

Source	Provider	Resolution	Freshness	Best for
NASA GIBS	VIIRS / MODIS	~250 m—1 km	~1 day + archive	Snow, smoke, floods; wide-area & historical
Sentinel-2	ESA Copernicus	~10 m	~5-day revisit	Water bodies, cities, NDVI, change
NASA FIRMS	VIIRS thermal	point data	~hourly	Active fire hotspots
Esri World Imagery	Mosaic	sub-meter	undated	Finest spatial detail
OpenStreetMap	Nominatim	—	—	Geocoding: place → bounding box

Trade-off: fresh + wide-area (GIBS) vs. detailed but a few days old (Sentinel-2) vs. very fine but undated (Esri). No paid satellite keys — all five are free and open. Agent figures out which one to plan and use and also iterates over next options if current selection is not useful.

Extension: from answer to deep-research report

Any query can be promoted into a longer, cited report — useful when a one-paragraph answer isn't enough.

How a report is built

- 1 Take the query (or pick one from history)
- 2 Re-run the agent with a longer, structured brief
- 3 Assemble sections + embed the imagery it used
- 4 Render to markdown in the app and to PDF



What's in the report

- Multiple sections written by the agent
- The actual satellite images it analysed, embedded
- Dates, sources, and stated limitations
- Saved to a Reports tab + linked from history
- Downloadable as a PDF



LIVE DEMO

Three questions, live against the engine

1

"Has it snowed at Lake Tahoe in the last few days — and if not, when did it last?"

Recent + historical imagery; the agent reasons across dates, not just "right now."

2

"How has Lake Mead changed since 2016?"

Change detection — two dated scenes, the before/after slider on the map.

3

"Was Lake Lagunita at Stanford filled with water at least once in the last ~5 years?"

Historical search over time on a small, local feature — relatable and specific.

Evaluation

To check it actually works, I built a small benchmark: 9 prompts across environmental, consumer, and finance questions, each run through the full agent (real tool calls) on two models, then graded 1–5 by an LLM judge on tool selection, groundedness, correctness, and clarity.

Overall scores (mean / 5)

Claude Opus 4.8

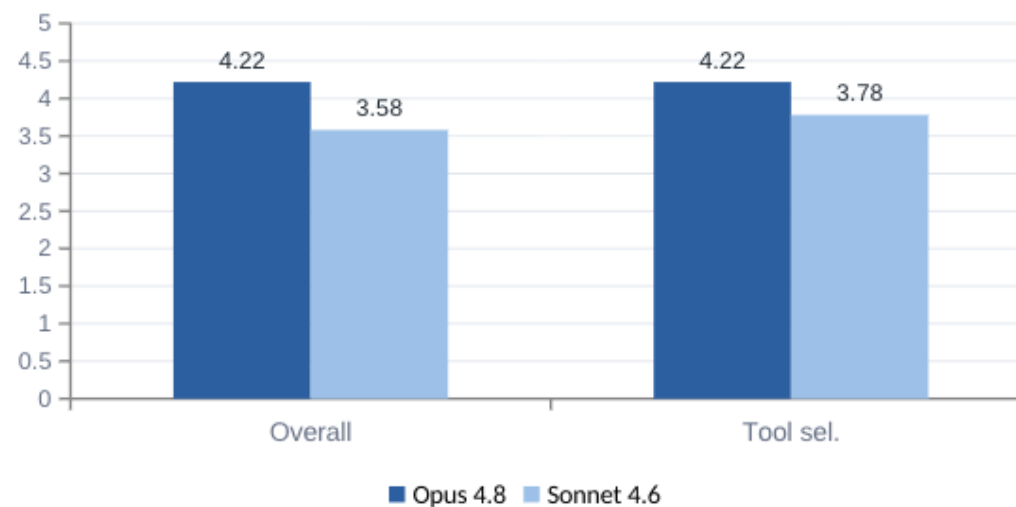
claude-opus-4-8

4.22 /5 33.7s avg

Claude Sonnet 4.6

claude-sonnet-4-6

3.58 /5 29.7s avg



Who it's for

The same engine serves very different users — from someone planning a trip to an analyst watching a supply chain.



Everyday & consumer

- Trip & conditions planning (snow, water)
- "What's happening at this place now?"
- General curiosity about anywhere



Environmental & public sector

- Wildfire & smoke monitoring
- Drought & reservoir levels
- Deforestation; disaster response



Research & journalism

- Fast, grounded fact-checks
- Visual evidence with dates & sources
- Reproducible, citable reports



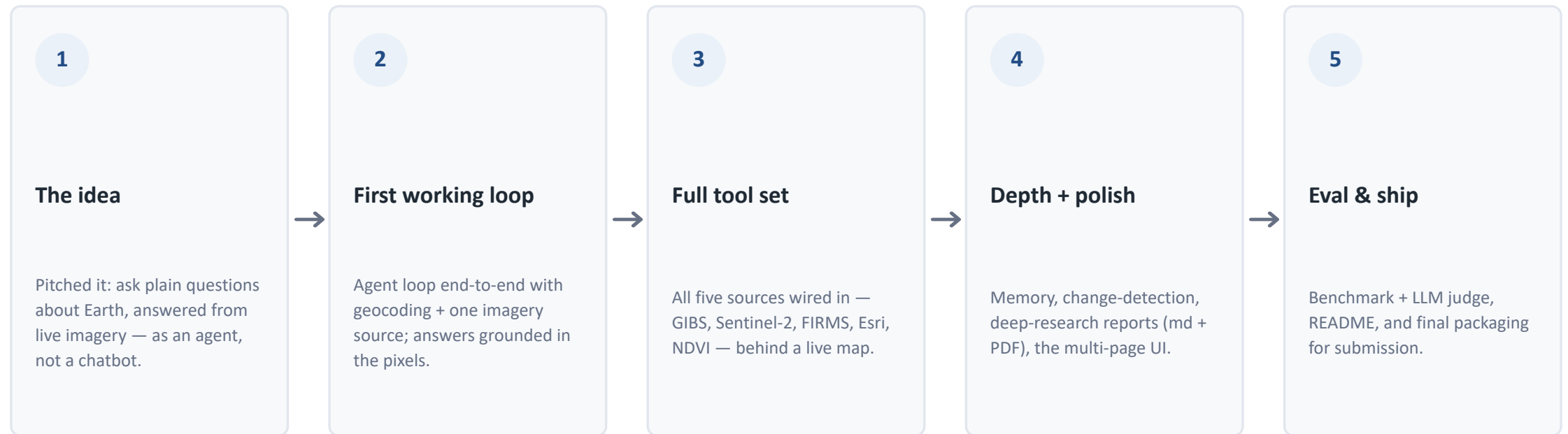
Commercial & finance

- Crop health / yield signals (NDVI)
- Storage & industrial activity over time
- Port / supply-chain monitoring

The consumer side is the widest audience; the commercial side ("satellite alpha") is where the data has the most measurable value.

How it came together

Each phase added a capability on top of a working core.



What I'd add next



More satellite sources

Sentinel-1 SAR to see through clouds and at night, deeper Landsat history, weather and ship layers.



RL post-training

Fine-tune the agent on its best benchmarked trajectories to sharpen tool selection and grounding.



News & language fallbacks

Cross-check imagery against local news; fall back to text when a scene is cloudy or unavailable.



Stronger evaluation & reporting

Human-verified labels, more models, repeated runs, an external judge; richer report templates and exports.



Product deployment and monitoring

Place-monitoring alerts, latency caching, accounts and saved workspaces, mobile.



Live, vision-grounded answers about anywhere on Earth — built on open satellite data and evaluated against a small benchmark.



github.com/rosikand/cs153-final-project

Code, setup, AI-usage disclosure, and citations are in the README.